

Mohamed Ezzat

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WORK EXPERIENCE

• Valeo

Software Engineer

Cairo, Egypt

Jan 2025 – Present

- Designed and implemented AI-driven agents integrating LLMs and Retrieval-Augmented Generation (RAG) to automate root cause analysis on build logs, providing developers with actionable fixes and reducing MTTR.
- Architected end-to-end CI/CD and CT (Continuous Testing) pipelines for powertrain systems delivered to major automotive OEMs, enforcing regulatory-compliant validation checks (e.g., ASPICE/ISO).
- Architected a high-throughput observability stack using OpenTelemetry, Loki, and Grafana, centralizing distributed pipeline traces to reduce build troubleshooting time by 70%.
- Engineered end-to-end data ingestion pipelines in Python to stream telemetry and validation data into Google BigQuery, leveraging SQL to query datasets and reducing report analysis time by over 60%.
- Engineered modular Ansible roles to automate infrastructure provisioning across 800+ remote machines, achieving 100% configuration parity and eliminating configuration drift.
- Developed custom automation tools and APIs in Go, packaging and deploying analytical applications using Docker and Kubernetes to ensure consistent runtime environments.

• Valeo

Embedded HiL Testing Engineer

Cairo, Egypt

Aug 2022 – Jan 2025

- Developed and executed HiL validation test cases for DC-DC converters and OBCs for major automotive clients including Ford, BMW, and Daimler, using dSPACE and Vector CANoe.
- Identified, documented, and tracked system-level bugs using Polarion ALM, collaborating with development teams to ensure timely resolution and verification of fixes.
- Analyzed system and software requirements to create and update test specifications, enhancing test coverage and validation effectiveness for new product features.
- Reduced bug resolution time by 40% through direct collaboration with developers.

EDUCATION

• Georgia Institute of Technology

Master of Science in Computer Science

Atlanta, USA

Expected Fall 2028

• Misr International University

B.Sc. in Electronics and Communication Engineering – Cum. GPA: 3.59

Cairo, Egypt

Graduated July 2022

- **Graduation Project:** Led the design and implementation of a satellite communication system, developing communication management software using AX.25 and SSP protocols.

SKILLS

- **Artificial Intelligence:** GenAI (LLMs, RAG, Agent Frameworks)
- **DevOps & Cloud:** Jenkins, Ansible, Docker, Kubernetes, ArgoCD, OpenTelemetry, Loki, Grafana, AWS, CI/CD
- **Programming & Data:** Python, Go, SQL, Groovy, C

CERTIFICATIONS & COURSES

- **Full Embedded Systems Diploma:** Covered C programming, Data Structures, AVR microcontrollers, RTOS, and Embedded Software Engineering Principles (Oct 2021).
- **Satellite Communication Subsystem Course:** Implemented the AX.25 protocol in LabVIEW and built a wireless data transmission system using Arduino and nRF modules (Sept 2021).